ASHWIN T E

Robotics Enthusiast | Embedded Systems Developer
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SUMMARY

An eager Embedded System Developer passionate about expanding expertise in electronics and autonomous systems. Recognized for efficiency in research and development, currently serving as the Programming Lead for Team MOVIS (2025–2026), a special team focusing on Space Technology, and the R&D Lead at the VIT Chennai Open Source Programming Club. Proven success in national and international robotics competitions.

TECHNICAL SKILLS

- **Systems & Robotics:** Embedded Systems, Robotics Operating System (ROS2), SLAM, Path Planning, Control Systems, PCB Modeling, CAD, CAM, CIM etc.
- **Design & Simulation:** Computer Aided Design (CAD), SolidWorks, LTSpice, Fusion 360,
- **Programming:** Embedded C, Python, C, C++, JAVA, Assembly, React, Tailwind, SQL.
- **Hardware:** Arduino, Raspberry Pi 5, Jetson Nano, ESP32, STM32.

EXPERIENCE & INTERNSHIPS

- **MEL Systems and Services | Intern | May 2025 – June 2025**
  Completed a 60-day professional internship focusing on electronic systems and hardware services.
- **VIT Chennai Open Source Programming Club | R&D Lead | 2025 – Present**
  Leading research initiatives and achieving impressive outcomes through efficient project management.

PROJECTS

- **ISRO Robotics Challenge | Jan 2025 – Present**
  Developing an autonomous drone utilizing ROS2, Raspberry Pi, and Pixhawk. Responsible for motor control and autonomous flight logic.
- **Cold Plasma Effects on Plant Growth (Patent Pending) | Programming Lead**
  Leading the software development and automation systems for research into plasma-assisted agricultural growth.
- **Autonomous Delivery Robot | Mar 2025 – Present**
  Developing an industrial-grade robot with obstacle detection and SLAM capabilities. Designed the electronics architecture including custom PDB and PCB.

ACHIEVEMENTS & AWARDS

- **NASA Human Exploration Rover Challenge (HERC) 2026:** Finalist in the RC subdivision, securing 6th place on the podium. Ranked as the Best Indian Team and the Fastest Team (7 minutes and 2 seconds). Handled electrical implementation as an Programming Lead for Team MOVIS (a Space Technology focused team).
- **Caterpillar Autonomy Challenge (IITM & Caterpillar):** Semi-finalist; specialized in autonomous navigation.
- **REVIVE 2024:** Winner of the National Level Hackathon.
- **Innovate National Hackathon:** Secured 3rd Place.
- **Project Expo (BITS Hyderabad):** Finalist (Reached the final round).
- **Smart India Hackathon (SIH) 2023:** Waitlisted placement.

EDUCATION

- **Bachelor of Technology in Mechatronics and Automation**
  Vellore Institute of Technology (VIT), Chennai | July 2024 – Present

ADDITIONAL INFORMATION

Languages: English, German, Tamil, Hindi.